

$$A) \quad O \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad A \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} \quad B \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix} \quad C \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$1) S_{\text{пар}} = \left| \frac{1}{6} \vec{OA} \vec{OB} \vec{OC} \right| = \left| \frac{1}{6} \det \begin{pmatrix} 2 & 3 & 1 \\ 5 & 5 & 2 \\ 2 & 2 & 1 \end{pmatrix} \right| = \frac{1}{6} |2 \cdot 1 - 3 \cdot 1 + 1 \cdot 0| = \frac{1}{6}$$

$$2) \quad (\vec{OA} \vec{OB} \vec{OC}) < 0 \Rightarrow \text{они не коллинеарны} \Rightarrow \text{есть базис } \{0, \vec{OA}, \vec{OB}, \vec{OC}\}$$

$\Downarrow$
базис левый

$$\begin{cases} \vec{OA}; \vec{OB}; \vec{OC} \end{cases} \vec{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{cases} e_1; e_2; e_3 \end{cases} \vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 5 & 2 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x + 3y + z \\ 5x + 5y + 2z \\ x + 2y + z \end{pmatrix}$$

$$\begin{cases} 1 = 2x + 3y + z & (1) \\ 2 = 5x + 5y + 2z & (2) \\ 3 = x + 2y + z & (3) \end{cases}$$

$$\begin{cases} 4 = 3x + 5y + 2z & (1) + (3) \\ 2 = 5x + 5y + 2z & (2) \end{cases}$$

$$-2 = 2x$$

$$\underline{x = -1} \quad (1) - (2) \quad -2 = x + y \quad y = -1$$

$$\underline{z = 6}$$

$$\in \{0; \vec{OA}; \vec{OB}; \vec{OC}\} \vec{a} = \begin{pmatrix} -1 \\ -1 \\ 6 \end{pmatrix}$$

$$3) \quad S = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 5 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

$$4) \quad [\vec{BC}, \vec{OB}] = \det \begin{pmatrix} e_1 & e_2 & e_3 \\ 2 & -3 & -1 \\ 3 & 5 & 2 \end{pmatrix} = -e_1 + e_2 - e_3$$

$$BC = \begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}$$

$$[\vec{BC}, \vec{OB}] = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

5) $\Pi_P \vec{OA}$ на (OBC)

$$OA' = \vec{OA} - \frac{(\vec{OA}, \vec{n}_{(OBC)})}{|\vec{n}_{(OBC)}|^2} \cdot \vec{n}_{(OBC)}$$

$$\vec{n}_{(OBC)} = [\vec{OB}, \vec{OC}] = \begin{vmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ 3 & 5 & 2 \\ 1 & 2 & 1 \end{vmatrix} = \vec{e}_1 - \vec{e}_2 + \vec{e}_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$\vec{OA} \cdot \vec{n} = 2 - 5 + 2 = -1$$

$$|\vec{n}|^2 = 1^2 + (-1)^2 + 1^2 = 3$$

$$OA' = \begin{pmatrix} 2 \\ 5 \\ 2 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{7}{3} \\ \frac{14}{3} \\ \frac{7}{3} \end{pmatrix}$$

6) $\vec{OB} = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}$
 $\vec{CA} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \quad (\vec{OB}, \vec{CA}) = 3 + 15 + 2 = 20$

$$|\vec{OB}| = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$|\vec{CA}| = \sqrt{1 + 9 + 1} = \sqrt{11}$$

$$(\vec{OB}, \vec{CA}) = |\vec{OB}| |\vec{CA}| \cdot \cos \alpha \quad \alpha - \text{угол } |\vec{OB}| |\vec{CA}|$$

$$\cos \alpha = \frac{20}{\sqrt{38} \sqrt{11}}$$

$$\alpha = \arccos \frac{20}{\sqrt{418}}$$

B)

$$\begin{pmatrix} 1 & 5 \\ 21 & 1 \\ -17 & 13 \end{pmatrix} \begin{pmatrix} 2 & -2 & 3 \\ 11 & 7 & -4 \end{pmatrix} = \begin{pmatrix} 2+55 & -2+35 & 3-20 \\ 42+11 & -42+7 & 63-4 \\ -34+143 & 134+137 & -17 \cdot 3 - 134 \end{pmatrix} = \begin{pmatrix} 57 & 33 & -17 \\ 53 & -35 & 59 \\ 109 & 125 & -103 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -2 & 3 \\ 11 & 7 & -4 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 21 & 1 \\ -17 & 13 \end{pmatrix} = \begin{pmatrix} 2-42-51 & 10-2+39 \\ 11+147+68 & 55+7-52 \end{pmatrix} = \begin{pmatrix} -91 & 47 \\ 226 & 10 \end{pmatrix}$$